Synoptic Question 2: Geothermal Power Generation

Reading Passage

Read the following article and then answer the questions that follow.

Geothermal power is a renewable energy source that harnesses the heat from the Earth's interior. This heat originates from the original formation of the planet and from the radioactive decay of materials in the Earth's core. In a geothermal power plant, this thermal energy is used to generate electricity.

One type of geothermal plant, a "flash steam" plant, pumps hot water under high pressure from deep underground reservoirs to the surface. The temperature of this water can be over 180 °C. When the water reaches the surface, the pressure is suddenly reduced, causing some of the water to rapidly vaporize, or "flash," into high-pressure steam.

This steam is then used to spin a turbine, which is connected to a generator to produce electricity. The remaining water and condensed steam are then injected back into the reservoir to be reheated, making it a sustainable resource.

Consider a geothermal plant that pumps hot water from a reservoir 2.5 km below the surface. The water enters the plant at 182 °C and a pressure of 8.0 x $10^5 Pa. After flashing, the steam enters the turbine at the same temperat$

The Earth's internal heat is partly due to the decay of radioactive isotopes, such as Uranium-238. A typical granite rock has a Uranium-238 concentration of 3 parts per million by mass. Geologists use seismic waves to locate these underground reservoirs.

Data:

- Depth of reservoir: 2.5 km
- Temperature of water from reservoir: 182 °C
- Specific heat capacity of water: 4200 J kg^{-1} K^{-1}
- Specific latent heat of vaporization of water (at 8.0 x $10^5 Pa$) : $2.05x10^6 J/kg$
- Electrical power output of the plant: 55 MW
- Efficiency of thermal to electrical energy conversion: 15
- Mass of Uranium-238 atom: 3.95×10^{-25} kg
- Half-life of Uranium-238: 4.47×10^9 years
- Energy released per decay of a U-238 atom: 4.27 MeV
- Speed of P-waves (sound) in rock: 5500 m s^{-1}

Questions

(a) Calculate the work done against gravity to lift 1 kg of water from the reservoir to the surface.

(b) The process of flashing involves a phase change.

  (i) Explain in terms of molecular motion why energy is required to change a substance from a liquid to a gas at a constant temperature.

  (ii) When 1.0 kg of the 182 °C water is brought to the surface, only a fraction of it flashes to steam. The energy for this vaporization comes from the cooling of the entire mass of water. If the water and steam are in thermal equilibrium at 175 °C after flashing, calculate the mass of steam produced from 1.0 kg of hot water.

(c) The steam from the flashing process drives the turbine.

  (i) Calculate the rate at which thermal energy must be supplied by the steam to the turbine.

  (ii) Using your answer from (b)(ii), or otherwise, estimate the mass of hot water that must be pumped from the reservoir each second.

(d) Briefly describe how the rotational motion of the turbine leads to the generation of an alternating electromotive force (e.m.f.) in the generator.

(e) A significant portion of the Earth's internal heat comes from the radioactive decay of Uranium-238.

(i) Calculate the decay constant, λ, for Uranium-238 in s^{-1}.

(ii) Calculate the activity of 1.0 kg of granite rock, assuming all the radioactivity is from the decay of Uranium-238.

(f) Suggest one environmental advantage of geothermal power over fossil fuel power plants.

(g) Geologists use seismic P-waves (a type of sound wave) to find the depth of reservoirs. A P-wave is sent from the surface, reflects off the rock layer above the reservoir, and returns to a detector.

(i) If the reservoir is 2.5 km deep, calculate the time for a P-wave to travel to the reservoir and back to the surface.

(ii) P-waves are longitudinal waves. Describe what is meant by a longitudinal wave.

(h) Not all the thermal energy supplied to the turbine is converted into electrical energy.

(i) State the principle of conservation of energy.

(ii) Using this principle and your answer to (c)(i), calculate the rate at which waste heat is expelled from the power plant.

Solutions

(a) Work done $= mgh = (1)(9.81)(2500) = 24525$ J

(b) (i) In a liquid, molecules are held by intermolecular forces. Energy is required to overcome these forces and separate the molecules, moving them far apart as a gas. This increases the potential energy of the molecules, not their kinetic energy, so the temperature is constant during the phase change.

(ii) Let m_s be the mass of steam.
Energy lost by cooling water = Energy gained to form steam.
$(1.0 \text{ kg}) \times c_{water} \times \Delta T = m_s \times L_v$
$(1.0)(4200)(182 - 175) = m_s \times (2.05 \times 10^6)$
$29400 = m_s \times (2.05 \times 10^6)$
$m_s = \frac{29400}{2.05 \times 10^6} = 0.0143$ kg

(c) (i) Rate of thermal energy input $= \frac{\text{Electrical Power Output}}{\text{Efficiency}} = \frac{55 \times 10^6}{0.15} = 3.67 \times 10^8$ W

(ii) From (b)(ii), 1.0 kg of water produces 0.0143 kg of steam. The thermal energy for this comes from the water cooling. So, the thermal energy extracted per kg of water is 29400 J. Mass of water per second $= \frac{\text{Total thermal energy needed per second}}{\text{Energy extracted per kg of water}} = \frac{3.67 \times 10^8}{29400} = 12480$ kg s^{-1}.

(d) The turbine spins a coil of wire within a magnetic field. This causes a continuous change in the magnetic flux linkage with the coil. By Faraday's law of induction, this induces an e.m.f. The continuous rotation naturally produces an alternating (sinusoidal) e.m.f.

(e) (i) $T_{1/2} = 4.47 \times 10^9$ years $= (4.47 \times 10^9) \times (365.25 \times 24 \times 3600) = 1.41 \times 10^{17}$ s.
$\lambda = \frac{\ln(2)}{T_{1/2}} = \frac{0.693}{1.41 \times 10^{17}} = 4.91 \times 10^{-18}$ s^{-1}

(ii) Mass of U-238 in 1 kg of granite $= 3 \times 10^{-6}$ kg.
Number of U-238 atoms, $N = \frac{\text{mass}}{\text{atomic mass}} = \frac{3 \times 10^{-6}}{3.95 \times 10^{-25}} = 7.59 \times 10^{18}$.
Activity, $A = \lambda N = (4.91 \times 10^{-18})(7.59 \times 10^{18}) = 37.3$ Bq

(f) Geothermal power plants do not burn fossil fuels, so they produce negligible greenhouse gas emissions (like CO_2), which contribute to climate change.

(g) (i) Total distance $= 2 \times$ depth $= 2 \times 2500 = 5000$ m. Time $= \frac{\text{distance}}{\text{speed}} = \frac{5000}{5500} = 0.91$ s.

(ii) A longitudinal wave is one in which the oscillations of the particles of the medium are parallel to the direction of energy transfer.

(h) (i) The principle of conservation of energy states that energy cannot be created or destroyed, only transformed from one form to another.

(ii) Rate of Heat Expelled = (Rate of Thermal Energy Input) - (Rate of Useful Electrical Work).
Rate of Heat Expelled $= (3.67 \times 10^8 \text{ W}) - (55 \times 10^6 \text{ W}) = 3.12 \times 10^8$ W.

[Image of a Flash Steam Geothermal Power Plant]

Figure 1: Diagram of a Flash Steam Geothermal Power Plant

CIE Physics 9702 Paper 4 Style

Variant 2 (Higher Difficulty)

Geothermal Power Generation — Open-Ended Extension

Instructions:

- Answer all parts. Show full working and state assumptions clearly.
- Use an appropriate number of significant figures.
- For open-ended parts, credit is awarded for valid physics reasoning, not only one final value.

Data and constants

- Reservoir depth $d = 2.5 \times 10^3\,\mathrm{m}$
- Temperature of water at wellhead $T_{\mathrm{in}} = 182\,^\circ\mathrm{C}$
- Post-flash equilibrium temperature $T_{\mathrm{f}} = 175\,^\circ\mathrm{C}$
- Specific heat capacity of water $c = 4.20 \times 10^3\,\mathrm{J\,kg^{-1}\,K^{-1}}$
- Latent heat of vaporisation at operating pressure $L_v = 2.05 \times 10^6\,\mathrm{J\,kg^{-1}}$
- Electrical output power $P_e = 55\,\mathrm{MW}$
- Turbine-generator efficiency (thermal to electrical) $\eta = 0.15$
- Rock density (for this question) $\rho_r = 2.7 \times 10^3\,\mathrm{kg\,m^{-3}}$
- Specific heat capacity of rock $c_r = 790\,\mathrm{J\,kg^{-1}\,K^{-1}}$
- $g = 9.81\,\mathrm{m\,s^{-2}}$

Question 1

The plant operator proposes to increase electrical output from 55 MW to 80 MW while keeping the same reservoir and flash process.

(a) Calculate the minimum thermal power input required at 80 MW, assuming efficiency is unchanged. [2]

(b) Using an energy-balance argument for flash steam production, show that the steam mass fraction from 1.0 kg of hot water is approximately 1.43×10^{-2}. [3]

(c) Hence estimate the required mass flow rate of hot water for

  (i) the current 55 MW plant,

  (ii) the proposed 80 MW plant.

  Comment on one engineering implication of your result. [4]

(d) **Open-ended design physics:** The current production well has internal radius 0.18 m.

  (i) Estimate the average upward speed of water for the 80 MW case.

  (ii) Discuss whether this speed is physically realistic for long-term operation.

  Your discussion must include at least two distinct physics considerations. [6]

Question 2

The separated brine and condensed steam are re-injected into hot rock. Assume a simplified thermal store of rock volume $V = 1.8 \times 10^8\,\mathrm{m}^3$ initially at $260\,^\circ\mathrm{C}$. A safety policy limits average reservoir cooling to $12\,\mathrm{K}$.

(a) Estimate the maximum thermal energy that may be extracted before reaching this cooling limit. [3]

(b) Using your result from Question 1(c)(ii), estimate an operating lifetime (in years) before the cooling limit is reached for the 80 MW plan, neglecting natural reheating. [4]

(c) **Open-ended evaluation:** Explain why this model may either under-estimate or over-estimate real reservoir lifetime. Include at least three well-developed physics points. [6]

Question 3

Geophysicists map the reservoir using P-wave reflections. A pulse is transmitted vertically and returns after $0.92\,\mathrm{s}$.

(a) Estimate the wave speed in the overlying rock. [2]

(b) A second survey later records return time $0.95\,\mathrm{s}$ along nearly the same path. Suggest two possible physical reasons for the change, and for each reason explain the direction of effect on wave speed. [4]

(c) **Open-ended methods question:** Propose one additional measurement (other than P-wave travel time) that would improve confidence in estimating reservoir condition. Explain clearly how your chosen measurement links to the relevant physics. [4]

Sample Solution (Variant 2)

Q1(a)

$$P_{\text{th},80} = \frac{P_e}{\eta} = \frac{80 \times 10^6}{0.15} = 5.33 \times 10^8\,\text{W}$$

Q1(b) For 1.0 kg water cooling from 182°C to 175°C,

$$Q = mc\Delta T = (1.0)(4.20 \times 10^3)(7) = 2.94 \times 10^4\,\text{J}$$

If this energy vaporises mass m_s:

$$m_s L_v = Q \Rightarrow m_s = \frac{2.94 \times 10^4}{2.05 \times 10^6} = 1.43 \times 10^{-2}\,\text{kg}$$

Q1(c) Thermal energy extracted per kg hot water is $2.94 \times 10^4\,\text{J}\,\text{kg}^{-1}$.

$$\dot{m} = \frac{P_{\text{th}}}{2.94 \times 10^4}$$

At 55 MW: $P_{\text{th}} = 55/0.15 = 3.67 \times 10^8\,\text{W}$,

$$\dot{m}_{55} = 1.25 \times 10^4\,\text{kg}\,\text{s}^{-1}$$

At 80 MW: $P_{\text{th}} = 5.33 \times 10^8\,\text{W}$,

$$\dot{m}_{80} = 1.81 \times 10^4\,\text{kg}\,\text{s}^{-1}$$

One implication: substantially higher pumping demand increases parasitic power and mechanical stress, reducing net efficiency margin.

Q1(d) Cross-sectional area:

$$A = \pi r^2 = \pi(0.18)^2 = 1.02 \times 10^{-1}\,\text{m}^2$$

Take water density $\rho_w \approx 10^3\,\text{kg}\,\text{m}^{-3}$.

$$v = \frac{\dot{m}}{\rho_w A} = \frac{1.81 \times 10^4}{(10^3)(1.02 \times 10^{-1})} \approx 1.78 \times 10^2\,\text{m}\,\text{s}^{-1}$$

This is very high. Realistically, two valid concerns include: large frictional pressure losses ($\propto v^2$), and severe pipe erosion/cavitation risk in two-phase regions. Additional valid points include pump power penalty, vibration, and reduced well lifetime.

Q2(a)

$$E_{\max} = \rho_r V c_r \Delta T = (2.7 \times 10^3)(1.8 \times 10^8)(790)(12) = 4.61 \times 10^{15}\,\text{J}$$

Q2(b) At 80 MW, thermal extraction rate $P_{\text{th}} = 5.33 \times 10^8\,\text{W}$.

$$t = \frac{E_{\max}}{P_{\text{th}}} = \frac{4.61 \times 10^{15}}{5.33 \times 10^8} = 8.65 \times 10^6\,\text{s} \approx 1.00 \times 10^2\,\text{days} \approx 0.27\,\text{years}$$

Q2(c) Any three developed points, e.g.: natural conductive/convective reheating extends life; extracted energy is not spatially uniform so local cooling can be faster; porosity and fluid pathways evolve with stress and mineral deposition, changing heat-transfer rates; reinjection temperature and placement can either preserve or degrade useful thermal gradients.

Q3(a) Round-trip distance $= 2d = 5000\,\text{m}$, so

$$v_p = \frac{5000}{0.92} = 5.43 \times 10^3\,\text{m}\,\text{s}^{-1}$$

Q3(b) Two acceptable reasons with explanation, e.g.: increased rock temperature lowers elastic modulus, reducing v_p and increasing return time; microcrack growth or fluid saturation changes effective bulk/shear moduli and density, typically reducing speed if cracking dominates.

Q3(c) Example: monitor attenuation (Q factor) or reflected amplitude spectrum. Increased cracking/fluid phase change modifies scattering and absorption, so combining travel time with attenuation separates geometric depth effects from rock-property changes.

Mark Scheme (CIE P4 style, Variant 2)

Total: 38 marks

Part	Answer / marking points	Marks
1(a)	$P_{\text{th}} = P_e/\eta$ (M1), substitution and correct value 5.33×10^8 W (A1).	2
1(b)	Uses $mc\Delta T$ with $\Delta T = 7$ K (M1), equates to $m_s L_v$ (M1), obtains $m_s \approx 1.43 \times 10^{-2}$ kg (A1).	3
1(c)(i)	Correct method from thermal power and energy per kg (M1), $\dot{m}_{55} \approx 1.25 \times 10^4$ kg s^{-1} (A1).	2
1(c)(ii)	$\dot{m}_{80} \approx 1.81 \times 10^4$ kg s^{-1} with correct scaling or full calculation (M1A1).	2
1(c) comment	One valid engineering implication linked to increased flow (B1).	1
1(d)(i)	Computes area and speed method (M1), speed around 1.8×10^2 m s^{-1} (A1).	2
1(d)(ii)	Any two developed physics considerations (2+2): e.g. pressure loss, cavitation/boiling, erosion, pump power, vibration/material limits.	4
2(a)	Uses $E = \rho V c \Delta T$ (M1), correct substitution (M1), $E_{\text{max}} \approx 4.6 \times 10^{15}$ J (A1).	3
2(b)	Uses $t = E/P$ (M1), obtains about 8.6×10^6 s (A1), converts to sensible year value ≈ 0.27 y (A1), clear interpretation (B1).	4
2(c)	Open-ended: up to 6 marks for three explained points (2 marks each). Award 1 mark for a valid point, +1 for clear physics linkage.	6
3(a)	Uses round-trip distance with $t = 0.92$ s (M1), $v_p \approx 5.4 \times 10^3$ m s^{-1} (A1).	2
3(b)	Two physically valid reasons with direction of effect explained (2 marks each).	4
3(c)	Proposes one relevant independent measurement (B1), explains link to reservoir physics (B1), explains why it improves confidence/complements travel time (B1), coherent scientific communication (B1).	4
	Total	**38**